



How to get an asymptotic of the real-valued branch of the inverse function?

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Consider function $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, defined as $f(x) = x + x^2 (1 + \log x)$. I need to find an asymptotic approximation of its inverse function $f^{(-1)} : \mathbb{R}^+ \rightarrow \mathbb{R}^+$, satisfying $f^{(-1)}(x + x^2 (1 + \log x)) = x$, for $x \rightarrow 0^+$. I made a few attempts:



```
AsymptoticSolve[x + x^2 (1 + Log[x]) == y, {x}, y -> 0, Direction -> -1,
Assumptions -> y > 0, SeriesTermGoal -> 2]
```



This returns unevaluated (I'm not sure if I use it correctly). Let's try to use `InverseFunction` :

```
Series[InverseFunction[Function[x, x + x^2 (1 + Log[x])]][y], y -> 0,
Assumptions -> y > 0, SeriesTermGoal -> 2]

(* E^(-1 + ProductLog[-1, -E]) + y/(-1 + E^(-1 + ProductLog[-1, -E])) -
((3 + 2 ProductLog[-1, -E]) y^2)/(2 (-1 + E^(-1 + ProductLog[-1, -E]))^3) +
0[y]^3 *)
```

This is better. It's a correct asymptotic, but for a non-real branch of the inverse function. I'm interested in the real-valued branch. If we replace `Series` with `Asymptotic`, we get the same result, just without $0[y]^3$ at the end.

Let's try to explicitly say the function is only defined for real arguments:

```
Series[InverseFunction[Function[x, ConditionalExpression[x + x^2 (1 + Log[x]),
Im[x] == 0]]][y], y -> 0,
Assumptions -> y > 0, SeriesTermGoal -> 2]

(* InverseFunction[Function[x, ConditionalExpression[x + x^2 (1 + Log[x]), Im[x]
== 0]]][0] +
Undefined y + Undefined y^2 + 0[y]^3 *)
```

Not helpful. If we replace `Series` with `Asymptotic`, we just get plain `Undefined`.

```
InverseFunction[Function[x, ConditionalExpression[x + x^2 (1 + Log[x]), Im[x] ==
0]]][y] + 0[y]^3

(* InverseFunction[Function[x, ConditionalExpression[x + x^2 (1 + Log[x]), Im[x]
```

```
== 0]]][0] +
      Undefined y + Undefined y^2 + 0[y]^3 *)
```

The same.

```
InverseSeries[x + x^2 (1 + Log[x]) + 0[x]^3, y]
```

It returns unevaluated.

How do I get an asymptotic of the real-valued branch of $f^{(-1)}(y)$?

Update: An example from the documentation for [ConditionalExpression](#) that shows how to use it with [InverseFunction](#), which I tried to follow:

Inverse of a function with a restricted domain:

```
In[1]:= InverseFunction[ConditionalExpression[#^2 + # + 1, # ≤ -1/2] &]
```

Out[1]=
$$-\frac{1}{2} - \frac{1}{2} \sqrt{-3 + 4 \#1} \quad \text{if } \#1 > \frac{3}{4} \quad \&$$

Update: Another class of elementary functions in asymptotic of whose inverses I'm interested in is similar to this:

```
Asymptotic[InverseFunction[ConditionalExpression[# + #^Sqrt[2], # >= 0] &][z], z
-> 0,
Assumptions -> z > 0, SeriesTermGoal -> 4]
```

Expected result:

$$(x \mapsto x + x^{\sqrt{2}})^{(-1)}(z) = z - z^{\sqrt{2}} + \sqrt{2} z^{2\sqrt{2}-1} - \frac{6-\sqrt{2}}{2} z^{3\sqrt{2}-2} + \mathcal{O}(z^{4\sqrt{2}-3})$$

equation-solving

symbolic

series-expansion

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edited Dec 13, 2020 at 23:08

asked Dec 11, 2020 at 22:55



Vladimir Reshetnikov

6,651 32 83

After comparing coefficients manually, it appears that the asymptotic is $y - y^2 (1 + \text{Log}[y]) + 0[y]^3$, but I still need to automate this process, because my actual function is more complicated than

the example I included in my question, and I may need to compute more terms.

– Vladimir Reshetnikov Dec 12, 2020 at 4:02



Although I upvoted your question, I guess you should reformulate it more precisely.

`InverseFunction` should be evaluated on `ConditionalExpression` like e.g. `InverseFunction[ConditionalExpression[#^3 + 1, # > 0] &][x]` while you seem to evaluate `ConditionalExpression` under a pure function. It's natural to expect greater flexibility of functions in `Asymptotic` and perhaps some improvements will appear in next versions of the system. Nonetheless one can find e.g. `Asymptotic[x + x^2 (1 + Log[x]), x -> 0]` or `Limit[1 + x (1 + Log[x]), x -> 0, Direction -> "FromAbove"]`. – Artes Dec 12, 2020 at 11:40



One can find that even `AsymptoticEqual[InverseFunction[InverseFunction[# + #^2 (1 + Log[#]) &]][x], x, x -> 0]` yields `True`. – Artes Dec 12, 2020 at 11:43

@Artes Do you mean there is a difference between `Function[ConditionalExpression[#1 + #1^2 (1 + Log[#1]), Im[#1] == 0]]` and `Function[x, ConditionalExpression[x + x^2 (1 + Log[x]), Im[x] == 0]]`? I was under impression that they are completely equivalent, except that the latter uses named parameter `x` instead of anonymous positional parameter `#1`.

– Vladimir Reshetnikov Dec 12, 2020 at 21:11



I would also expect they are equivalent however one can observe with simple examples this is not the case here. Nevertheless in my comment I've meant you should not confuse an appropriate order of evaluation, namely it should be `ConditionalExpression[PureFunction, ...]` and not just `PureFunction[ConditionalExpression[...]]`. – Artes Dec 12, 2020 at 22:16

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